

WASP-104b

R-2546

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_180224 · created 2026-08-26T00:33:15+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458173.8066 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.043
Transit depth (Rp/Rs)^2	2.22 +/- 1.27 [%]
Orbital Inclination (inc)	84.44 +/- 2.81
Airmass coefficient 1 (a1)	0.117 +/- 0.086
Airmass coefficient 2 (a2)	1.91 +/- 1.49
Scatter in the residuals of the lightcurve fit is	79.0 %
Best Comparison Star	#1 - [423, 27]
Optimal Aperture	2.41
Optimal Annulus	8.54
Transit Duration (day)	0.061 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.149 $\pm$ 0.043 vs 0.1214 (deviation 0.64 σ)
Duration fit vs published	0.061 d vs 0.0742 d (deviation 0.69 σ)
Mid-time O–C	1.7 min
Depth SNR	3.5
Residual scatter	79.0 %

Light curve

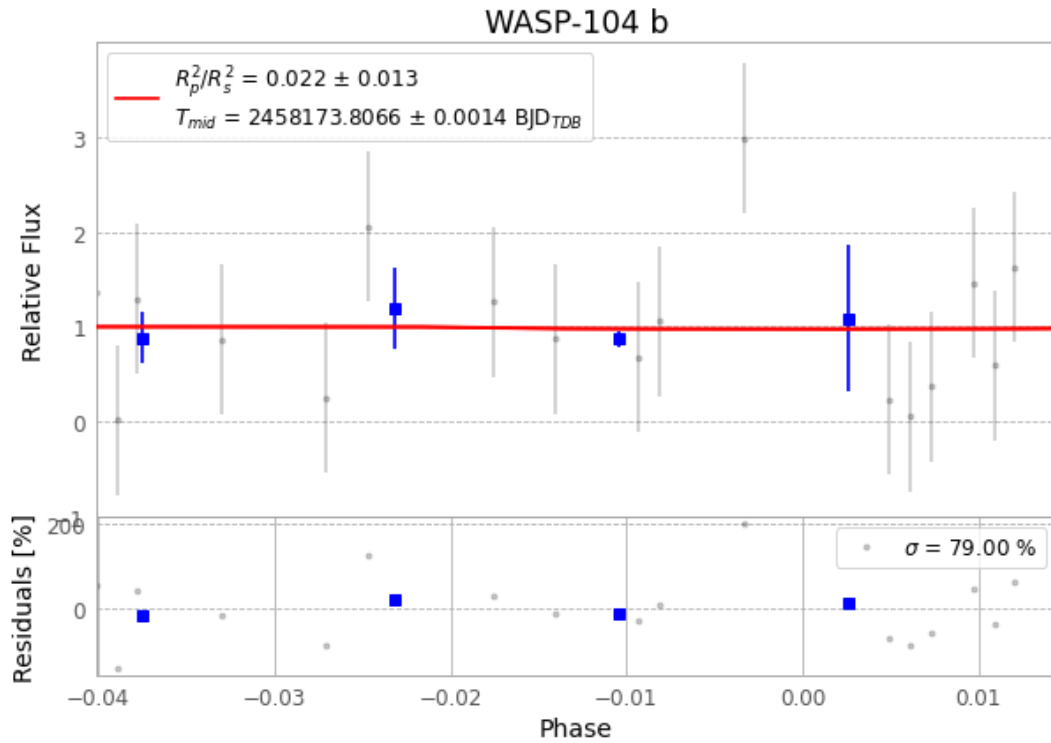


Figure 1 — FinalLightCurve_WASP-104 b_2018-02-23.png (SHA-256 d740c9d3d7a0e51f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [445, 195], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 82bf6be946025faf...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

50 FITS frames (33 MB), WASP-104180224052412.FITS → WASP-104180224075112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-180224.log (f7020d12f9fa...), FinalLightCurve_WASP-104 b_2018-02-23.png (d740c9d3d7a0...), FinalLightCurve_WASP-104 b_2018-02-23.csv (ef6baa079d4c...)

Compute resources

Wall time 184.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 00:33Z.